PROJECT PLAN

Lifecycle, Reuse vs Build, WBS, Schedule and Tools

Project: Smart Expense & Budget Management Application

Group Number: G-15

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Appendix E: Project Plan

E.1 Lifecycle Justification

Agile (Scrum-style sprints) fits this project because requirements around budgeting and reporting features are likely to evolve as user feedback comes in during demos. Short, fixed-length sprints let the team deliver working increments early starting with login in Sprint 1, gather feedback, and adjust scope — e.g. deciding how deep to go on sandbox bank-linking without blocking the whole project on upfront design of every feature.

E.2 Reuse vs. Build Classification

Component	Classification	Notes
Django web framework	Reuse	Off-the-shelf framework, not built by the team
Django's built-in authentication (password hashing, sessions)	Reuse	Used as-is for FR-001–004, SEC-001
Chart.js	Reuse	Off-the-shelf charting library for FR-017 and FR-018
HTMX	Reuse	Off-the-shelf library for partial page updates
Django Admin (base)	Reuse (developer tool)	Built-in admin site kept for developers only, at `/django-admin/` and limited to superusers; FR-020 to FR-022 are delivered by team-built, role-protected views in the categories app
Accounts app (registration/profile logic)	Build	Custom Django app, team-written
Transactions app (expense/income + CSV import)	Build	Custom Django app, team-written
Budgets app (budget tracking + alerts)	Build	Custom Django app, team-written
Recurring app (recurring transaction scheduler)	Build	Custom Django app, team-written
Reports app (analytics/dashboard queries)	Build	Custom Django app, team-written
Categories app (category master list, role-protected admin views)	Build	Custom Django app, team-written

E.3 Work Breakdown Structure

WP	Tasks	Owner	Est. Effort (person-days)	Output
WP1	Requirements baseline + use cases	All	4	SRS
WP2	Architecture, ER model, UML	All	4	Architecture/ Design
WP3	Auth & profile module	Sarthki Vachhani (SV)	6	Working feature
WP4	Expense/income transaction module	Sai Manogna BV (SM)	7	Working feature
WP5	Budget creation & tracking module	Rhea Sushanth (RS)	6	Working feature
WP6	Reports & analytics dashboard	Rhea Sushanth (RS)	6	Working feature
WP7	Alerts, recurring transactions (incl. scheduler-triggered generation, FR-024), admin panel, sandbox bank integration (FR-023, stretch)	SV / SM / RS	7	Working feature
WP8	Testing, CI/CD, security hardening	All*	6	Test/validation evidence
WP9	Final validation & report	All	4	Final report

**Testing can be split among the team.*

E.4 Sprint Schedule / Gantt Chart

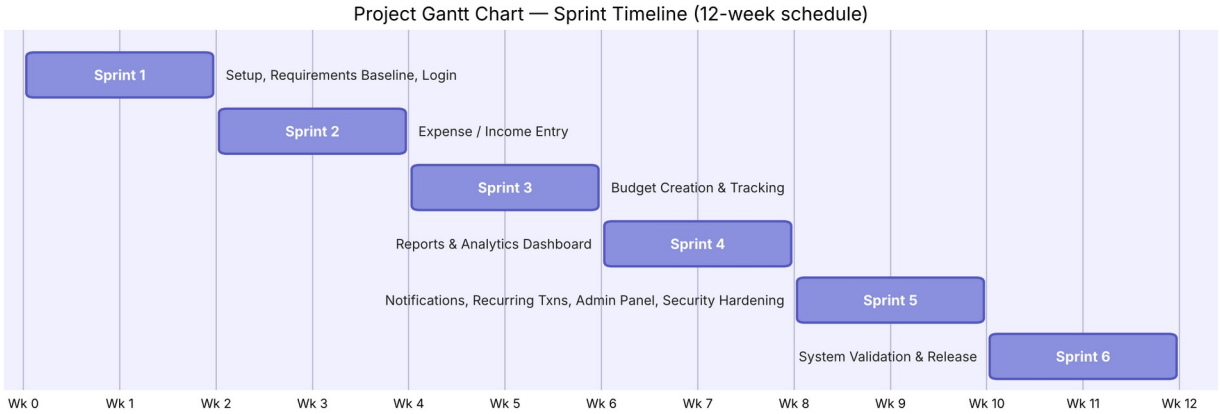


Figure 4: Project Gantt chart showing the 6-sprint, 12-week schedule (sprints run back to back: weeks 0-2, 2-4, 4-6, 6-8, 8-10, 10-12)

E.5 Tool Stack

Activity	**Recommended / Example Tool**	**Evidence to Submit**
Backend	Python (Django)	Repository code, running app
Database	PostgreSQL / SQLite (dev)	Schema/migration files
Frontend	Django templates + HTMX + Chart.js	Screens, chart demo
Version control	Git + GitHub	Repository URL, commit history
Agile planning	GitHub Projects	Backlog, user stories, sprint board
CI	GitHub Actions	Pipeline run log
Testing	Pytest / Django test framework	Test results
API/UI testing	Postman / Django test client	Collection/results
Design	PlantUML / Mermaid.js	UML / ER diagrams